

Intergenerational Housing, Fertility, and Demographic Transition

Theory and quantitative architecture

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Abstract

Young households need additional space when they have children, but access to family-sized housing depends on tenure, liquid wealth, and the housing retained by older owners. I study the resulting feedback between fertility and the housing market. A two-generation model shows that reproductive stationarity selects the long-run housing cost, while market clearing selects population scale. It also gives exact conditions under which reproduction can be below replacement while household population and current housing costs continue to rise: a large inherited cohort must add more housing demand as it ages than is lost from the smaller cohort behind it. I then embed the mechanism in a lifecycle model with sequential births, income risk, saving, tenure choice, collateral constraints, child maturation, survival, and bequests. The calendar-time equilibrium propagates the unnormalized joint distribution and clears the housing market along the transition. A reproducible numerical illustration separates inherited demographic momentum from a no-fertility-decline path and compares policies that relax young access with policies that release housing retained by older households. Finally, an audit of the maintained quantitative specification shows what remains before the transition can be interpreted as a U.S. quantitative result: the existing fertility block is below replacement and responds only weakly to housing costs. The framework is therefore a coherent theoretical and computational design; estimating the U.S. transition remains a separate empirical and numerical task.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run incidence of housing policy.

The analysis proceeds in two layers. Section 2 makes the equilibrium logic transparent with young and old households. Sections 3–4 retain the paper’s lifecycle, tenure, and intergenerational structure. The numerical section then uses an age-structured version of the same accounting to show the transition. The maintained quantitative model is evaluated separately, so illustrative results are not confused with calibrated estimates.

2 Two-generation theory

2.1 Environment

Young households. At date t , a mass Y_t of young households has income y . A young household chooses consumption c , adult housing h , and children n . Preferences are

$$u(c, h, n) = c + \alpha \log h + \theta \log n, \quad \alpha > 0, \quad \theta > 0. \quad (1)$$

The parameter θ is the value of children. Quasi-linearity removes income effects and keeps the price mechanism available in closed form. Throughout this section, income is high enough that the interior allocation is feasible: $y > \alpha + \theta$.

Child costs and young access. Each child requires $q \geq 0$ units of ordinary goods and $a > 0$ units of housing. The market cost of that housing is ap , where p is the price of housing services. A goods-equivalent wedge $\omega \geq 0$ captures the additional cost of obtaining family space through a small rental market, a down payment, or another tenure constraint. The full cost of a child is therefore $q + a(p + \omega)$.

Old households. A mass O_t of old households no longer chooses fertility. After solving its retention and estate problem, an old household's housing demand is summarized by

$$d_O(p; \tau) = \frac{\varphi(\tau)}{p}, \quad \varphi'(\tau) > 0. \quad (2)$$

The parameter τ indexes forces that keep old households in housing, including transaction costs and the estate motive. The lifecycle model below derives retention from choices rather than imposing equation (2).

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H}p^\varepsilon, \quad \bar{H} > 0, \quad \varepsilon \geq 0. \quad (3)$$

The fixed-stock case is $\varepsilon = 0$. The number $\bar{n} > 0$ is effective replacement fertility: after survival and household formation, a young household with $\bar{n}$ children produces one young household in the next generation.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t , from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c, h, n} c + \alpha \log h + \theta \log n \quad \text{subject to} \quad c + ph + [q + a(p + \omega)]n = y. \quad (4)$$

At an interior solution, child demand and total housing demand are

$$n_Y(p; \theta, \omega) = \frac{\theta}{q + a(p + \omega)}, \quad d_Y(p; \theta, \omega) = \frac{\alpha}{p} + an_Y(p; \theta, \omega). \quad (5)$$

A higher market price and a tighter access wedge have the same sign in the child decision. They differ in equilibrium incidence: the market price clears housing, while the access wedge changes which households can obtain it.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^\varepsilon = Y_t d_Y(p_t; \theta, \omega) + O_t d_O(p_t; \tau). \quad (6)$$

For inherited cohort masses, the left side rises and the right side falls with p_t , so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1} = \mathcal{R}(p_t; \theta, \omega) Y_t, \quad O_{t+1} = Y_t, \quad \mathcal{R}(p; \theta, \omega) = \frac{n_Y(p; \theta, \omega)}{\bar{n}}. \quad (7)$$

The function $\mathcal{R}$ is effective reproduction. It is strictly decreasing in the housing cost and the young-access wedge.

Definition 1 (Reproductive steady state). A reproductive steady state is a price $p^* > 0$, positive cohort masses (Y^*, O^*) , and household choices such that households optimize, housing clears, and $(Y_{t+1}, O_{t+1}) = (Y_t, O_t)$.

Proposition 1 (Steady state and policy incidence). *A unique positive reproductive steady state exists if and only if $\theta > \bar{n}(q + a\omega)$. It is*

$$p^* = \frac{\theta/\bar{n} - q}{a} - \omega, \quad Y^* = O^* = \frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha + \varphi(\tau) + a\bar{n}p^*}. \quad (8)$$

A lower value of children θ or a tighter young-access wedge ω lowers both the long-run housing cost and household population. Stronger old retention raises $\varphi(\tau)$ and lowers population without changing the reproductive housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and leaves the reproductive housing cost unchanged.

Reproduction and housing clearing perform different jobs. Replacement selects the housing cost at which a cohort reproduces itself. Housing clearing then selects how many young and old households can occupy the market at that cost. This triangular ordering is why permanent housing supply is capitalized into population rather than the long-run price in the closed economy.

The existence condition is economically sharp. Even if market housing were free, a child would still cost $q + a\omega$. If $\theta \leq \bar{n}(q + a\omega)$, cheaper market housing alone cannot restore replacement.

2.4 Demographic momentum

The steady-state result does not imply that population and prices fall immediately. Define the old-to-young ratio $\mu_t = O_t/Y_t$ and current effective reproduction $\mathcal{R}_t = \mathcal{R}(p_t; \theta, \omega)$. The next result compares the smaller young cohort created by low fertility with the inherited cohort that is currently moving into old age.

Proposition 2 (Low fertility with rising population and housing costs). *Suppose the preference, access, retention, and supply primitives are unchanged between t and $t + 1$, and $\mathcal{R}_t < 1$. Household population rises between t and $t + 1$ if and only if*

$$\mu_t < \mathcal{R}_t < 1. \quad (9)$$

The housing cost rises if and only if

$$(1 - \mu_t)d_O(p_t; \tau) > (1 - \mathcal{R}_t)d_Y(p_t; \theta, \omega). \quad (10)$$

Hence low fertility, population growth, and housing-cost growth coexist exactly when

$$\mu_t < \min \left\{ \mathcal{R}_t, 1 - (1 - \mathcal{R}_t) \frac{d_Y(p_t; \theta, \omega)}{d_O(p_t; \tau)} \right\}. \quad (11)$$

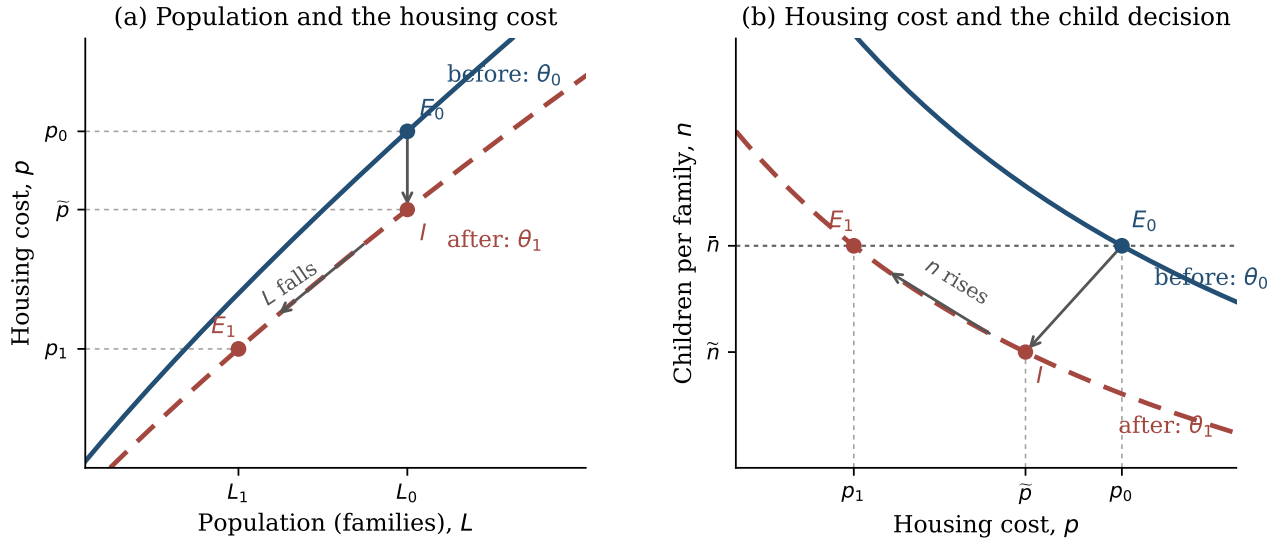


Figure 1: Reproduction selects the steady-state housing cost, and housing clearing maps that cost into population. After a fall in the value of children, fertility is initially below replacement. A lower population then reduces the price and partly offsets the original shock.

Equation (10) is an accounting test. The smaller next young cohort removes $(1 - \mathcal{R}_t)Y_t d_Y$ units of demand. The unusually large current young cohort adds $(Y_t - O_t)d_O$ units as it ages. The housing cost rises when the second force is larger. From an exact stationary age distribution, $\mu_t = 1$, so both population and the housing cost fall after a fertility decline. A positive transition requires an inherited cohort bulge generated by earlier fertility, immigration, or both.

The fertility decline nevertheless lowers housing costs relative to the no-shock path. Demographic momentum can make the realized cost rise over time; it cannot reverse the counterfactual effect of the shock.

2.5 Intergenerational policy

Proposition 1 distinguishes three interventions. Relaxing the young-access wedge ω raises fertility at a given cost and supports a higher reproductive housing cost and population. Reducing old retention τ releases housing, lowers the transition cost, and raises fertility during adjustment; in the new closed-economy steady state it raises population while the reproductive housing cost returns to the level selected by θ and ω . Expanding housing supply has the same long-run housing-cost neutrality but does not directly change which generation occupies the stock.

This is where the demographic closure strengthens the paper's original intergenerational mechanism. Reallocating one home toward a constrained young household can affect its current tenure and fertility, and that birth changes the size of the future entrant cohort. The second effect is absent when the adult population is normalized after every solution.

3 Lifecycle environment

The quantitative model retains the mechanism but replaces the representative young and old households with finite lives, sequential births, income risk, saving, discrete ownership, continuous rental housing, survival, and bequests. It is a one-market model: the focus is intergenerational allocation and tenure, not spatial sorting across locations.

3.1 States and timing

One model period is four years. Thus the household problem uses $\beta = \beta_{\text{annual}}^4$ and a four-year gross liquid return R_b ; maintenance, property taxes, and other flow rates are converted to the same period before entering a budget. A household enters adult life at age 18 and can live through age 85. Its beginning-of-period state is

$$x = (a, b, \xi, o, n, \mathbf{c}) \in \mathcal{X}. \quad (12)$$

Here a is age, b is liquid net wealth, ξ is persistent earnings, o is the inherited tenure and owner-size state, $n \in \{0, 1, 2, 3+\}$ is children ever born, and $\mathbf{c}$ is the vector of dependent-child ages or maturation stages. Let $m(\mathbf{c})$ be the number currently at home. Tracking the stages separately prevents a later birth from resetting the maturation clock of an older sibling. The retained E5b code instead uses one shared child clock, so $\mathbf{c}$ is a proposed timing repair, not merely a new population closure. It changes entrant timing and requires re-estimation before quantitative use. Conditional on household continuation, each child's stage advances independently. The separate population pipeline introduced below preserves the child's future-entry claim even if the parental household exits.

The within-period order is: the household observes fertility shocks and, if eligible, chooses whether to attempt the next birth; it chooses whether to rent or own and, if owning, a house size; it chooses consumption, rental space, and next-period liquid wealth; income and child states then evolve, survival is realized, and surviving households age. A successful birth increases parity by one. Children who mature generate potential adult entrants rather than being added mechanically to their parent's household state.

3.2 Preferences, children, and bequests

Let $m = m(\mathbf{c})$, let $e(m)$ be an equivalence scale, and let $\alpha(m)$ be the consumption share when m children are at home. The maintained sequential specification uses

$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad \alpha(m) = \alpha_0 - \Delta_0 \mathbf{1}\{m > 0\} - \Delta_1 m. \quad (13)$$

The first object raises total household needs; the second shifts expenditure toward housing when children are present. For consumption c and housing services s , flow utility is

$$u(c, s; m) = e(m)^{\sigma-1} \frac{\left[c^{\alpha(m)} s^{1-\alpha(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m, \quad \sigma > 1. \quad (14)$$

Rental services equal the chosen quantity h^R . An owner of physical size H_j receives services χH_j , where χ is the owner-service premium. Physical rooms, not utility services, enter market clearing.

At death, liquid wealth and net sale proceeds from housing form estate wealth w^e . The maintained child-invariant warm glow is

$$\mathcal{B}(w^e) = \theta_0 \frac{(\theta_1 + \max\{w^e, 0\})^{1-\sigma}}{1-\sigma}. \quad (15)$$

Bequests affect saving and old-age housing retention. They do not determine a particular descendant's entry wealth; the model is intergenerational in housing allocation and cohort accounting, not fully dynastic. The displayed utility is deliberately not recentered at zero wealth because the maintained benchmark is not normalized; recentering would change fertility value levels and would require re-estimation.

3.3 Income, tenure, and housing finance

Working income is $y(a, \xi)$, with a lifecycle profile and a persistent Markov component. Retirees receive a pension. Liquid wealth earns gross return R_b . Renters choose $h^R \in [0, \bar{h}^R]$. Owners choose from the physical menu $\{H_1, \dots, H_J\}$.

Let P_t be the asset price of one unit of housing and r_t its rent. A seller pays proportional cost ψ^s . If the household changes from inherited owner size $H(o)$ to new size $H(o')$, the transaction changes liquid resources by

$$\mathcal{T}_t(o, o') = \mathbf{1}\{H(o') \neq H(o)\} [(1 - \psi^s)P_t H(o) - P_t H(o')]. \quad (16)$$

Here $H(R) = 0$. Keeping the same house creates no transaction. A new buyer must finance the down payment from cash:

$$\text{cash before purchase} \geq (1 - \phi)P_t H(o'), \quad (17)$$

where ϕ is the financed share.

The maintained post-purchase rule distinguishes secured capacity from an age-tapered unsecured position. Let

$$\underline{b}_t^C(o') = -\phi P_t H(o'), \quad \mathcal{U}_{a+1}(u) = \min\{s_{a+1}^d \min(u, 0), -D_{a+1}\}, \quad (18)$$

where s_{a+1}^d tapers existing unsecured debt toward zero and D_{a+1} is the age-specific credit line. A renter satisfies $b' \geq \mathcal{U}_{a+1}(b)$; an owner satisfies

$$b' \geq \underline{b}_t^C(o') + \mathcal{U}_{a+1}(b - \underline{b}_t^C(o')). \quad (19)$$

At the retained benchmark $D_{a+1} = 0$: existing unsecured debt can initially roll over but cannot increase, and the taper forces repayment later in life. This is the rule in the active solver; it is not a special no-deleveraging exception for incumbent owners.

Conditional on renting, the continuous choices satisfy

$$c + r_t h^R + b' = R_b b + y(a, \xi) + \mathcal{T}_t(o, R) + T_t, \quad 0 \leq h^R \leq \bar{h}^R. \quad (20)$$

Conditional on owner size $H(o') > 0$, they satisfy

$$c + (\delta_H + \tau_t^p)P_t H(o') + b' = R_b b + y(a, \xi) + \mathcal{T}_t(o, o') + T_t. \quad (21)$$

Mortgage interest is embedded in the return on negative liquid wealth. The transfer T_t is zero in the baseline and is determined by the government budget in funded policy experiments.

3.4 Sequential fertility

During the fertile ages, a household at parity $n < 3+$ chooses between waiting and attempting the next birth. A conception probability f_a maps an attempt into a birth. Additive Type-I extreme-value shocks have scale κ_1 on the first-birth margin and κ_C on continuation margins. Separating the two scales allows the model to match childlessness and completed fertility without interpreting a rooms response as a second-birth hazard. Each scale-specific shock is recentered to have mean zero. This normalization matters because κ_1 and κ_C differ.

Conditional on a price path, let $W_t(x)$ be the optimized value of the rest of the period conditional on entering it in state x . The value of waiting is $V_t^0(x) = W_t(x)$ and the value of attempting is $V_t^1(x) = f_a W_t(x^+) + (1 - f_a)W_t(x)$, where x^+ increments parity and adds a newborn child stage. The attempt probability is

$$\pi_t^F(x) = \frac{\exp\{V_t^1(x)/\kappa(n)\}}{\exp\{V_t^0(x)/\kappa(n)\} + \exp\{V_t^1(x)/\kappa(n)\}}, \quad \kappa(n) = \begin{cases} \kappa_1, & n = 0, \\ \kappa_C, & n \geq 1. \end{cases} \quad (22)$$

Under that mean-zero normalization, the beginning-of-period inclusive value is

$$V_t(x) = \kappa(n) \log \left[e^{V_t^0(x)/\kappa(n)} + e^{V_t^1(x)/\kappa(n)} \right]. \quad (23)$$

Fertility is therefore a flow hazard; children ever born is a stock. The calendar-time shifter used below changes the flow value of children and is not described as an identified causal preference shock.

3.5 The household problem

Let Q_t^ξ denote the earnings transition and s_a the probability of surviving to the next age. Let $\epsilon_t^H(o')$ be the tenure and owner-rung taste shock. Given the full future sequence of rents, asset prices, taxes, transfers, and child-demand shifters, the post-fertility value used above is

$$W_t(x) = \mathbb{E}_{\epsilon_t^H} \left[\max_{o', c, h^R, b'} \left\{ u_t(c, s; m) + \epsilon_t^H(o') + \beta s_a \mathbb{E}_t[V_{t+1}(x')] + \beta(1 - s_a) \mathcal{B}(w_{t+1}^e) \right\} \right], \quad (24)$$

subject to the appropriate budget, transaction, debt, and housing-menu constraints above. Its continuation expectation integrates earnings and independent child-stage transitions; conception is already integrated in V_t^1 . Equations (22)–(23) then integrate the fertility taste shock.

Equation (24) is deliberately written as a path problem. Along a transition, current purchases occur at P_t and future sales occur at future prices. Solving a stationary Bellman problem separately at each date would omit expected capital gains and is not a transition equilibrium.

4 Calendar-time equilibrium

4.1 Housing assets and construction

Housing is durable. Let K_t be the occupied housing stock and let I_t be net additions: gross construction minus physical retirements. Its law is

$$K_{t+1} = K_t + I_t, \quad I_t = \mathcal{I}(P_t, K_t; \zeta_t^S), \quad (25)$$

where ζ_t^S is a supply shifter. A fixed-stock transition sets $I_t = 0$. The numerical diagnostic below instead uses a static, price-elastic contemporaneous supply schedule and does not solve this stock law. I_t is a reduced-form net change in available stock. By convention there is no unit-level depreciation risk for a maintained incumbent house: it remains intact and can be resold.

Let $\delta_H P_t$ be maintenance that preserves an occupied unit. Competitive ownership of a rental unit then implies the deterministic asset-pricing condition

$$(R_b + \delta_H + \tau_t^p) P_t = r_t + P_{t+1}. \quad (26)$$

When the real asset price is constant, this reduces to $r = (R_b - 1 + \delta_H + \tau^p)P$, the maintained stationary user-cost relation. The full resale value follows from the no-unit-depreciation convention above. Transaction costs belong in household choices, not in the return condition of a frictionless marginal landlord.

4.2 The unnormalized cohort law

Let G_t be the finite measure of adult households over $\mathcal{X}$. Conditional on the equilibrium price path, household policies define a sub-Markov household continuation kernel $Q_t(x, A)$, expected births $f_t(x)$,

and current physical housing demand $h_t(x)$. Adult household continuation and child maturation are different processes, so the population closure also carries a child pipeline.

Let $C_{\ell t}$ be the mass of future adult-household equivalents that is ℓ periods from maturity, for $\ell = 1, \dots, L$. The conversion ν maps one birth into future household equivalents and ς_ℓ is child survival through stage ℓ . With births occurring at the end of a period,

$$\begin{aligned} F_t &= \int_{\mathcal{X}} f_t(x) dG_t(x), & B_{t+1} &= \varsigma_1 C_{1t}, \\ C_{\ell,t+1} &= \varsigma_{\ell+1} C_{\ell+1,t}, & \ell &= 1, \dots, L-1, & C_{L,t+1} &= \nu F_t. \end{aligned} \quad (27)$$

Thus B_{t+1} is the locally born household-formation pool. Because the pipeline is separate from G_t , a parental household exit does not erase a child's future-entry claim. The vector $\mathbf{c}$ still determines the parent's current needs; equation (27) handles aggregate demographic renewal. This aggregate closure assumes parental-household exit does not alter child survival or eventual household formation; modeling the child's interim guardian and housing demand would require an additional state.

Let $\mathcal{M}_{t+1}^E$ be an exogenous finite measure of outside entrants, let ρ_t be the fraction of locally born potential entrants retained in the market, and let Γ_{t+1}^B be their normalized entry-state distribution. Actual young-adult entry is

$$e_{t+1} = M_{t+1} + \rho_t B_{t+1}, \quad M_{t+1} = \mathcal{M}_{t+1}^E(\mathcal{X}). \quad (28)$$

The exact calendar law is

$$G_{t+1}(A) = \int_{\mathcal{X}} Q_t(x, A) dG_t(x) + \mathcal{M}_{t+1}^E(A) + \rho_t B_{t+1} \Gamma_{t+1}^B(A). \quad (29)$$

This equation carries cohort size, wealth, tenure, and children through time. It does not renormalize G_{t+1} . The current stationary forward routine constructs a whole normalized cross-section from an injected entrant cohort; a transition solver instead needs the one-period pushforward in equation (29).

Adult household population is

$$S_t = \int_{\mathcal{X}} dG_t(x). \quad (30)$$

Throughout the equilibrium analysis, “population” means adult-household mass. Total persons require a separate aggregation of adults and dependent children; the two series need not peak at the same date. In the baseline transition, $\{\mathcal{M}_t^E, \rho_t, \Gamma_t^B\}$ are specified exogenous paths. Endogenizing migration would require an outside value and a location-choice equation; it is an extension, not a missing condition in the baseline. For a national closed experiment, set $\mathcal{M}^E = 0$ and $\rho = 1$. For an empirical U.S. transition, immigrants should be distributed across entry ages rather than compressed into age 18 unless the compression is stated explicitly.

4.3 Market clearing

Let $h_t(x)$ be expected physical housing occupied after the current tenure and rental choices. Housing services clear at every date:

$$\int_{\mathcal{X}} h_t(x) dG_t(x) = K_t. \quad (31)$$

Together, equations (24), (25), (26), (29), and (31) close the calendar-time model.

Definition 2 (Transition equilibrium). Given an initial household distribution G_0 , child pipeline C_0 , housing stock K_0 , paths for outside entrant measures, local retention, locally born entry-state distributions, other exogenous shifters, and a terminal condition, a transition equilibrium is a sequence of values and household policies, distributions $\{G_t, C_t\}$, housing stocks and construction $\{K_t, I_t\}$, rents and asset prices $\{r_t, P_t\}$, and fiscal transfers such that households optimize, equations (25), (26), and (27)–(29) hold, housing clears by (31), and every policy budget balances.

The transition begins from the observed joint distribution, not an exact steady state. Matching only the initial total population is insufficient: demographic momentum depends on the joint age and tenure composition, while the young-access mechanism also depends on wealth and family status.

4.4 Stationary endpoints

The stationary endpoint can be written without ambiguity about normalization. At a constant price P , let Q_P^* be the household-continuation operator on measures and let γ_P be the normalized state distribution of one entrant cohort. Because adult life is finite, the lifetime adult measure produced by one new entrant household per period is

$$v_P = \sum_{j=0}^{J-1} (Q_P^*)^j \gamma_P = (I - Q_P^*)^{-1} \gamma_P. \quad (32)$$

Let $\bar{\varsigma} = \prod_{\ell=1}^L \varsigma_\ell$ be survival from birth to adult household formation. Define adult household-years per entrant and lifetime next-generation households by

$$\mathcal{A}(P) = \int_{\mathcal{X}} dv_P, \quad R_0(P) = \nu \bar{\varsigma} \int_{\mathcal{X}} f_P(x) dv_P(x). \quad (33)$$

The first object determines the stationary entry rate needed per adult; the second is the next-generation reproduction number. In an adult-normalized stationary cross-section,

$$E_0(P) = \frac{1}{\mathcal{A}(P)}, \quad B_0(P) = \frac{R_0(P)}{\mathcal{A}(P)}, \quad \boxed{\frac{B_0(P)}{E_0(P)} = R_0(P)}. \quad (34)$$

The three reproduction statistics used in this report are related but not interchangeable. $\mathcal{R}_t$ is next-cohort reproduction in the two-generation model, $R_0(P)$ is lifetime next-generation households per lifecycle entrant, and the frozen demographic eigenvalue below is a one-period asymptotic growth factor. Each equals one at its corresponding stationary replacement point, but their values away from replacement need not coincide.

Proposition 3 (Reproductive endpoint). *Suppose locally generated entrants enter with the same state distribution γ_P used in the renewal calculation. A nonzero closed-economy stationary measure exists at P^* if and only if*

$$R_0(P^*) = 1. \quad (35)$$

Conditional on this price, normalized composition is $g^ = v_{P^*}/\mathcal{A}(P^*)$. Housing clearing determines adult-household scale:*

$$S^* = \frac{K^s(P^*)}{\int_{\mathcal{X}} h(x; P^*) dg^*(x)}. \quad (36)$$

If R_0 is strictly decreasing, the reproductive price is unique.

The assumption on entrant states is substantive. Equation (34) is exact for the scalar renewal accounting used in the maintained model. It is also an exact representation of an unrestricted demographic operator only if descendants reproduce the wealth, earnings, and other state distribution used for entrants. The present bequest motive does not itself deliver that result.

The stationary housing stock in equation (36) is the solution $K^s(P)$ to $\mathcal{I}(P, K^s(P); \zeta^S) = 0$. Under a maintained fixed stock it is the exogenous constant $\bar{K}$.

Under the same common entrant-state distribution assumption, an outside entrant flow M and local retention ρ imply the stationary scale

$$S^* = \frac{M}{E_0(P) - \rho B_0(P)}, \quad E_0(P) > \rho B_0(P). \quad (37)$$

Housing clearing then selects P . The closed economy is the transparent theoretical benchmark. The open version is the empirically relevant regularization when reproduction is below replacement and immigration is positive. A national closure and a local housing market with migration are therefore different equilibrium objects.

5 Transition mechanics and interpretation

5.1 Exact adult-household momentum

Let $\sigma_t(x) = Q_t(x, \mathcal{X})$ be the probability that an adult household remains in the model next period, and let

$$D_t = \int_{\mathcal{X}} [1 - \sigma_t(x)] dG_t(x) \quad (38)$$

be adult-household exit. Integrating equation (29) gives the exact accounting identity

$$\boxed{S_{t+1} - S_t = e_{t+1} - D_t.} \quad (39)$$

Adult households grow when maturing locally born households plus immigrants exceed exits. Current births do not enter equation (39) until those children mature. Thus period fertility, frozen-environment reproduction, and current population growth are distinct objects.

This distinction resolves the apparent U.S. puzzle. Below-replacement fertility today is consistent with a growing adult population when the current age distribution reflects larger past birth cohorts or immigration. It is also consistent with rising housing demand when those cohorts are moving into ages with high ownership and space demand.

5.2 Exact housing-demand decomposition

Define expected next-period housing of a survivor by

$$\begin{aligned} \int_{\mathcal{X}} h_{t+1}(x') Q_t(x, dx') &= \sigma_t(x) \bar{h}_{t+1}^S(x), \\ e_{t+1} \bar{h}_{t+1}^{\text{new}} &= \int_{\mathcal{X}} h_{t+1}(x) d\mathcal{M}_{t+1}^E(x) \\ &\quad + \rho_t B_{t+1} \int_{\mathcal{X}} h_{t+1}(x) d\Gamma_{t+1}^B(x). \end{aligned} \quad (40)$$

Let $H_t^D = \int h_t(x) dG_t(x)$. The change in aggregate housing demand is

$$\begin{aligned} H_{t+1}^D - H_t^D &= \underbrace{\int \sigma_t(x) [\bar{h}_{t+1}^S(x) - h_t(x)] dG_t(x)}_{\text{aging, tenure transitions, and retention}} \\ &\quad - \underbrace{\int [1 - \sigma_t(x)] h_t(x) dG_t(x)}_{\text{housing released by exits}} + \underbrace{e_{t+1} \bar{h}_{t+1}^{\text{new}}}_{\text{new-household demand}}. \end{aligned} \quad (41)$$

Equation (41) is an exact quantity identity. In equilibrium its left side equals $K_{t+1} - K_t$, so its sign alone cannot predict the next rent. A valid price test must also respect asset no-arbitrage. Hold the continuation asset-price path $\mathcal{P}_{t+2}$ fixed. For a candidate rent r at date $t+1$, equation (26) implies

$$P_{t+1}(r) = \frac{r + P_{t+2}}{R_b + \delta_H + \tau_{t+1}^p}. \quad (42)$$

Let $\hat{h}_{t+1}(x; r, P_{t+1}(r), \mathcal{P}_{t+2:})$ be the housing policy under that candidate pair, and define

$$\hat{H}_{t+1}^D(r; \mathcal{P}_{t+2:}) = \int_{\mathcal{X}} \hat{h}_{t+1}(x; r, P_{t+1}(r), \mathcal{P}_{t+2:}) dG_{t+1}(x). \quad (43)$$

Applying the same three terms in equation (41), with every h_{t+1} replaced by $\hat{h}_{t+1}$, decomposes the shift in this demand schedule into survivor aging and tenure, housing released by exits, and new-household demand. The numerical packet computes the static-supply analogue of this fixed-cost decomposition.

Because K_{t+1} is predetermined by date- t investment, if next-period excess demand is strictly decreasing in its candidate rent, then

$$r_{t+1} > r_t \iff \hat{H}_{t+1}^D(r_t; \mathcal{P}_{t+2:}) > K_{t+1}. \quad (44)$$

This old-rent test is the full-model version of Proposition 2. It keeps the paper’s intergenerational mechanism visible: a large cohort can move from renting into ownership, retain family-sized homes in old age, and shift demand outward even as the entrant pipeline shrinks. The candidate asset price moves with the rent through equation (42); the full asset-price change still depends on the continuation path.

5.3 Where the transition starts

There need not be a single observed date at which a preference shock occurs. The empirical transition should start from the first date at which the joint state can be measured. Its initial condition must include the age–wealth–tenure–parity distribution, the children already in the maturation pipeline, and the housing stock. A smooth fertility wedge can then be inferred so that the model matches age-specific birth hazards over time. This wedge is a compact summary of omitted forces—childcare, work, partnership, norms, and other costs—not an identified causal preference shock.

The appropriate comparison is therefore between two paths with the same initial state. The baseline uses the inferred fertility-wedge sequence; the no-decline counterfactual freezes that wedge at its initial value. Low fertility can coexist with rising realized prices along the baseline because of inherited momentum. But, holding the other paths fixed, it should lower housing demand and prices relative to the no-decline counterfactual. This relative statement is the causal discipline the levels alone cannot provide.

5.4 Long-run policy incidence

At a regular closed-economy root, any policy g satisfies

$$\frac{dP^*}{dg} = -\frac{R_{0g}(P^*; g)}{R_{0P}(P^*; g)}. \quad (45)$$

If reproduction falls with the housing price, a policy that raises reproduction at a fixed price supports a higher reproductive price. Relaxing young access can have this effect because it changes tenure and fertility before children are born. Its long-run population effect combines the induced price with housing supply and per-household demand.

A pure old-release policy has different incidence. If it changes housing only after fertility and child maturation, and does not enter earlier continuation values, bequest utility, or taxes, then $R_{0g} = 0$. It leaves the reproductive price unchanged and accommodates more households by lowering housing per adult. This neutrality is not automatic in the full model: anticipated transaction costs and estate incentives can change earlier saving, tenure, and fertility decisions. The quantitative model must compute, rather than assume, the sign.

6 A reproducible numerical transition

The numerical exercise is deliberately small enough to audit. It demonstrates the cohort accounting and policy timing before a full solution of the lifecycle transition. It is not calibrated, not a forecast, and not a solved asset-price equilibrium.

6.1 Illustration

One period is five years. Adult households occupy twelve age groups from 20–24 through 75–79. Let x_{jt} be adult mass at age j , $q_{\ell t}$ a four-cohort child-maturation queue measured in future adult-household equivalents, σ_j adult survival, s_c child survival, and ν the household-formation equivalents generated by one birth. The demographic law is

$$x_{0,t+1} = s_c q_{3t}, \quad x_{j+1,t+1} = \sigma_j x_{jt}, \quad q_{0,t+1} = \nu \sum_j f_j(p_t, \theta_t, \omega_t) x_{jt}, \quad q_{\ell+1,t+1} = s_c q_{\ell t}. \quad (46)$$

Births occur at the end of period t and enter $q_{0,t+1}$ at the next date. The stages $q_0, \dots, q_3$ represent ages 0–4 through 15–19; the last stage enters x_0 at the following date. Thus adulthood occurs twenty years after the end-of-period birth even though it is five date indices after t . The fertility profile is concentrated around age 30. Its level falls smoothly from a replacement environment as θ_t declines. The initial age distribution has a prime-age bulge and smaller recent child cohorts. There is no post-initial immigration.

At each date, a contemporaneous housing-cost index clears

$$\bar{H}_s p_t^\varepsilon = \sum_j x_{jt} d_j(p_t; \theta_t, \omega_t, \tau_t). \quad (47)$$

Housing demand rises through midlife and remains high at older ages. The “frozen demographic growth factor” is the dominant eigenvalue of the adult aging, fertility, and maturation operator evaluated at current conditions. It is below one when the current environment is below replacement.

In the baseline, the frozen growth factor is below one after the initial date, yet both adult households and the housing-cost index rise over several five-year intervals. Adult households peak at 1.0325 and the housing cost at 1.0177, both in year 25 relative to initial values of one. By year 50, the indices have fallen to 0.9248 and 0.9464; by year 100 they are 0.7286 and 0.8159. These magnitudes have no empirical content. The result is the sign pattern: a declining endpoint can be preceded by decades of momentum.

The no-decline path starts from exactly the same inherited state. Its housing cost is 1.0316 in year 25 rather than 1.0177 in the baseline. By year 50, its adult-household and housing-cost indices are 0.9771 and 0.9907, versus 0.9248 and 0.9464 in the baseline. Thus inherited momentum explains why both realized paths can rise initially, while the gap between them isolates the effect of the post-initial fertility-schedule decline in this illustration.

A fertility decline with inherited demographic momentum

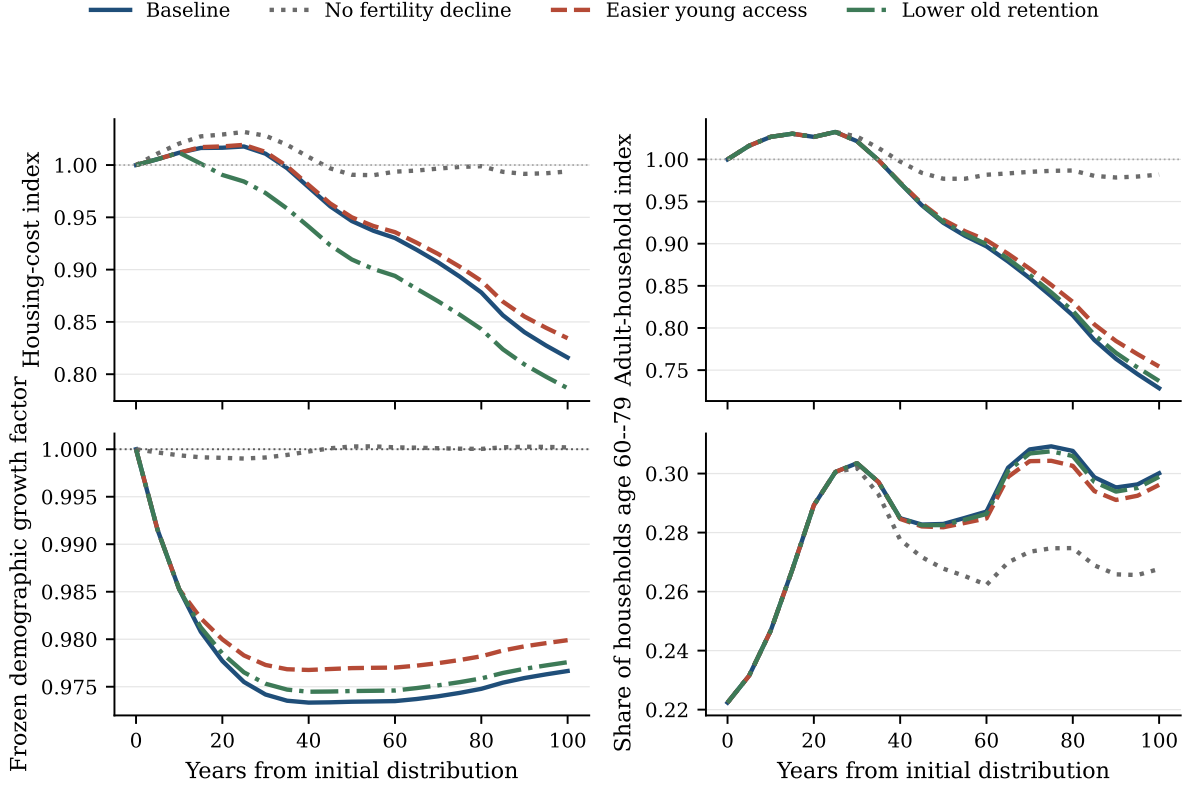


Figure 2: A generic fertility decline with inherited demographic momentum. All paths start from the same adult and child distributions. The no-decline path freezes the fertility schedule at its initial value. In the baseline, housing cost and adult-household mass rise initially even after the frozen demographic growth factor falls below one. Easier young access affects future entry after children mature; lower old retention releases housing immediately. All series are illustrative.

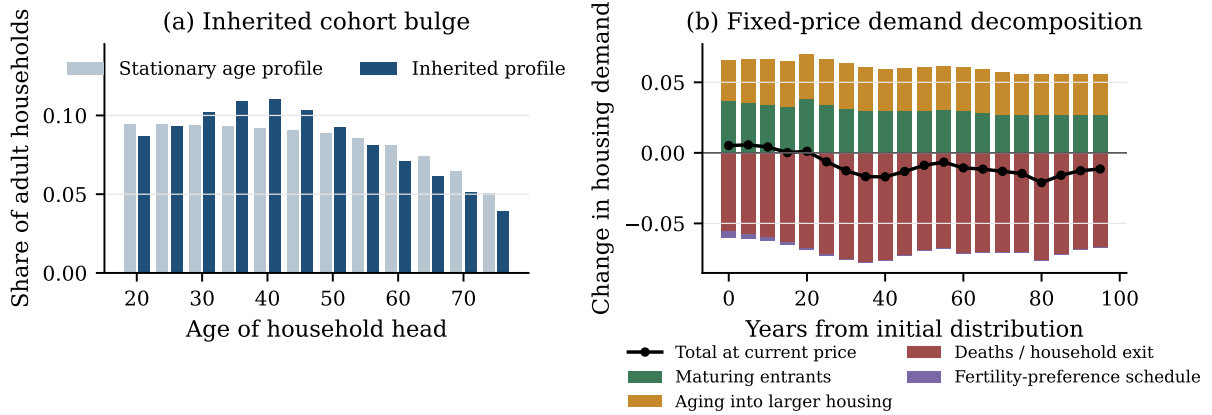


Figure 3: The inherited age profile and the exact fixed-price demand decomposition. Early in the transition, maturing entrants and aging into larger housing outweigh exits and the downward fertility schedule shift. Later, exits dominate.

The no-decline path keeps the value-of-children schedule at its initial level while preserving the same inherited age distribution and child pipeline. The easier-access experiment reduces the young-family access wedge after year 10. It raises fertility immediately but adult-household scale only after the

maturation delay. The lower-retention experiment reduces housing held at ages 60 and above. It lowers the clearing cost immediately and changes fertility indirectly through that cost. Neither experiment is a welfare calculation.

The numerical packet verifies housing clearing, adult accounting, and the demand decomposition to machine precision. Across all 80 nondegenerate scenario transitions, the fixed-price demand sign predicts the next clearing cost sign. The largest absolute housing residual, adult-accounting residual, and decomposition residual are respectively 4.44×10^{-16} , 2.64×10^{-16} , and 2.43×10^{-16} .

6.2 What the illustration does not do

The object p_t in equation (47) is a current housing-market clearing cost. It is not the forward-looking asset price P_t in the lifecycle model. Solving the latter requires backward household optimization and equation (26). The exercise also abstracts from income growth, interest-rate movements, construction, and immigration after the initial date. It proves feasibility of the mechanism; it does not explain the observed U.S. house-price path.

7 Mapping to the maintained quantitative model

7.1 What is retained and what is new

The intended dynamic model retains the maintained preferences and sequential birth margins: children shift equivalence-scaled needs toward housing; fertility is a sequence of birth decisions; renters face a size cap; buyers face the down-payment condition in equation (17); owners choose from a housing ladder; and transaction costs and warm-glow bequests generate old-age retention. It also makes one necessary architecture repair: independent child stages replace the retained shared clock. Beyond that repair, the dynamic closure adds the level distribution G_t , the aggregate child-maturation pipeline needed to form future entrants, and time-indexed values and prices. The retained benchmark parameters can therefore audit magnitudes and identification, but they cannot be carried into this transition without re-estimating the repaired fertility architecture.

This is not the one-shot completed-family-size model in the earlier circulated draft. That version remains useful for local analytical results, but it cannot be described as the same quantitative model as the retained equivalence-scale, sequential-birth specification. The present document uses the latter as the model that would be taken to the transition data.

The intergenerational emphasis is strengthened, not displaced. Young-access policies have an immediate tenure and fertility effect and a delayed population effect. Old-release policies have an immediate stock-allocation effect and an indirect fertility effect through prices. Reporting both components reveals which policies change family formation and which chiefly reallocate the existing housing stock.

In a final paper, the two-generation model can serve as the theory section and the lifecycle environment need be presented only once. The stationary benchmark establishes the household and intergenerational allocation channels; one added transition section then asks whether inherited cohorts reconcile low fertility with rising housing costs and how much demographic feedback amplifies policy.

7.2 Audited benchmark

The best retained July 25 E5b parameter vector has current-contract loss 385.8755, asset price 0.815625, and housing residual 1.11×10^{-6} . It is an audited benchmark, not a promoted calibration: the PSID rooms construction is under hold, and childlessness and old-wealth dispersion remain large failures.

Table 1 reports a few economically central rows; Appendix B reports the complete target system and every estimated or fixed parameter.

Table 1: Selected audited fit of the retained benchmark

Moment	Data	Model	Reading
Completed fertility	1.918	1.904	Close
Family ownership gap	0.168	0.166	Close
Rooms gap, 3+ versus 1–2 children	0.368	0.372	Close
Ownership, ages 30–55	0.575	0.564	Close
Childless share	0.188	0.080	Large miss
Old wealth p90/p50	3.448	2.068	Large miss

The benchmark requires entrant flow $E_0 = 0.0617335$ per adult and produces mature locally born flow $B_0 = 0.0511056$. Thus

$$\frac{B_0}{E_0} = 0.82784 < 1. \quad (48)$$

Treating the 3+ category with its reported top-code mean raises the diagnostic ratio to 0.90319, still below replacement. Across asset prices from 0.01 to two times the benchmark, the maintained ratio ranges only from about 0.836 to 0.820; its local price elasticity is -0.01067 . The maintained benchmark therefore has no positive closed reproductive root over the audited range.

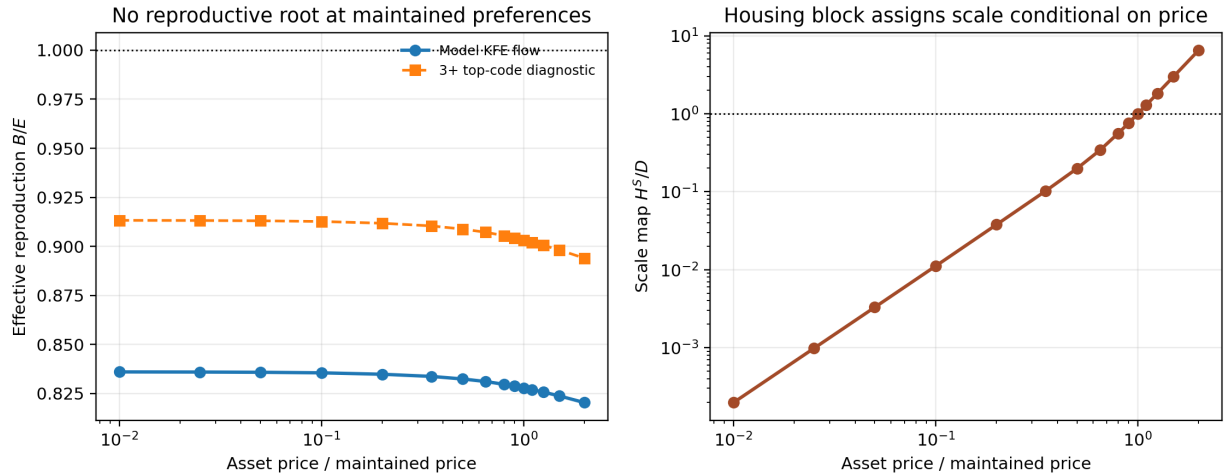


Figure 4: Reproductive and housing-scale maps at the retained benchmark. The left panel shows that neither the model flow nor the 3+ top-code diagnostic reaches replacement over the audited price range. The right panel shows that housing clearing nevertheless assigns stationary scale conditional on price.

The conclusion is architecture-sensitive. With the retained parameters but an independent maturation clock for each child, the audit gives $B_0/E_0 = 1.1179$; after re-estimating that architecture in the August 5 diagnostic, it gives 0.7852. Neither exercise is a valid promoted counterfactual: the former does not re-estimate behavior, and the latter uses a different fit. Their dispersion shows why child timing must be fixed before estimating a dynamic closure.

7.3 Identification consequence

Stationary fertility levels and timing observed at one market price do not identify the slope R_{0P} . The current target system contains strong moments for levels, timing, tenure, and rooms, but no causal housing-cost-to-birth hazard moment. The near-flat reproductive map in Figure 4 could therefore reflect either the model’s true economic restriction or an unidentified fertility-noise scale.

A quantitative transition needs a tenure-specific response moment that moves housing cost or access while holding other child costs fixed. The estimand should distinguish renters, prospective buyers, and incumbent owners because the same asset-price movement raises rent and down payments but also raises owner wealth. Without this slope evidence, the steady-state feedback and policy amplification should be presented as theory, not as an estimated magnitude.

8 Implementation and paper design

8.1 A minimal full-transition algorithm

The full transition can be implemented without redesigning the household problem.

1. Construct G_0 from the observed joint age, wealth, tenure, parity, and child-stage distribution. Construct K_0 from the housing stock. The initial child pipeline is required because it determines entrants over the first twenty years.
2. Specify paths for the reduced-form fertility wedge, outside entry, construction shifters, taxes, and any policy. The baseline wedge matches age-specific birth hazards; the no-decline counterfactual freezes it.
3. Guess paths for P_t and r_t . Given a terminal value function, solve household problems backward using equation (24).
4. Push G_t and the child pipeline forward one period using equations (27) and (29). Recompute births, entrants, exits, construction, and the period government budget.
5. Update rents and asset prices to satisfy housing clearing and dynamic no-arbitrage. Iterate backward and forward until every date clears and the terminal distribution is close to its terminal stationary endpoint.

The terminal endpoint can be open or closed. With the current benchmark it should remain open in reported numerical exercises: no usable closed reproductive root has been established over the audited price range. A changed and re-estimated fertility architecture could alter that conclusion. A period-by-period pension budget is also required: the stationary pay-as-you-go formula cannot be reused when the age distribution changes.

8.2 Validation gates

Before interpreting a computed transition, five gates should pass.

1. Independent child clocks and the conversion of the 3+ category must give consistent birth, maturation, and completed-fertility accounting.
2. The initial age distribution and child pipeline must reproduce observed near-term adult-household growth before prices are used to fit anything.
3. The baseline and no-decline paths must share G_0 , K_0 , migration, income, interest rates, and supply shifters; only the fertility wedge differs.
4. Every period must report housing demand, supply, residuals, rents, asset prices, entrants, exits, the quantity identity in equation (41), and the old-rent demand test in equation (44).
5. Policy experiments must balance their fiscal accounts and separate the immediate household response from the delayed demographic-scale response.

9 Conclusion

The model yields one clean separation. Reproductive balance selects a long-run housing cost in the closed economy; housing clearing selects population scale. The path between steady states is governed by inherited cohort composition, tenure transitions, old-age retention, immigration, and durable supply. It can therefore feature below-replacement reproduction, rising adult-household mass, and rising housing costs at the same time.

The simple theory isolates the mechanism, and the numerical illustration verifies its accounting. The lifecycle section provides a closed transition architecture once its stated exogenous entry and policy paths are supplied. What is not yet complete is a quantitative U.S. transition: the retained benchmark has no closed reproductive root, child maturation is architecture-sensitive, and the housing-cost elasticity of fertility is not identified by the current moments. Those facts define a focused empirical and computational agenda rather than a reason to abandon the closure.

The closest housing analogy is durable urban decline: as Glaeser and Gyourko (2005) emphasize, demand and prices can move well before the physical stock adjusts. The present mechanism adds the reverse link from housing costs to family formation and the cohort structure through which that link returns to future demand. The lifecycle fertility block is in the tradition of Sommer (2016) and Doepke and Kindermann (2019); the housing side builds on tenure and asset-market models such as Henderson and Ioannides (1983) and Sommer et al. (2013). The central synthesis is a calendar-time renewal equilibrium that keeps these channels in one accounting system; I do not claim that renewal theory itself is new.

A Proofs

Proof of Proposition 1. The first-order conditions in equation (4) give $h = \alpha/p$ and $n = \theta/[q + a(p + \omega)]$. Expenditures on adult housing and children are α and θ , so the stated income condition ensures positive consumption. Reproductive stationarity requires $n = \bar{n}$, which gives the price in equation (8). It is positive exactly when $\theta > \bar{n}(q + a\omega)$. In a steady state $Y = O$. Substituting young demand, old demand, and $n = \bar{n}$ into market clearing and multiplying by p gives the stated population. It is increasing in p : its log derivative is $(\varepsilon + 1)/p - a\bar{n}/[\alpha + \varphi(\tau) + a\bar{n}p] > 0$. The comparative statics follow directly. $\square$

Proof of Proposition 2. At date t , total household population is $(1 + \mu_t)Y_t$. At date $t + 1$ it is $(1 + \mathcal{R}_t)Y_t$, so it rises exactly when $\mathcal{R}_t > \mu_t$. Evaluate next-period housing demand at the current clearing price. Its change is

$$Y_t\{(\mathcal{R}_t - 1)d_Y(p_t) + (1 - \mu_t)d_O(p_t)\}.$$

Supply is unchanged and increasing and demand is decreasing in price. Therefore the next clearing price rises exactly when this expression is positive, which is equation (10). Combining the two strict inequalities gives equation (11). $\square$

Proof of Proposition 3. Let e be stationary entrant flow. Iterating the survivor law gives $G = ev_P$. The lifetime births generated by one entrant, multiplied by child survival and the household-formation conversion, equal $R_0(P)$. In a stationary child pipeline the mature flow therefore equals $eR_0(P)$: the pipeline delays a constant flow but does not change it. A nonzero closed stationary economy requires this flow to equal e , so $R_0(P) = 1$. Conversely, if $R_0(P) = 1$, any positive multiple of v_P reproduces its entrant flow. Normalizing gives $g = v_P/\mathcal{A}(P)$, and housing clearing fixes the multiple as in equation (36). $\square$

B Complete benchmark target fit

Table 2: Retained E5b benchmark: complete current-contract target system

Moment	Target	Model	Gap	Weight	Loss contribution
Completed fertility	1.918000	1.903573	−0.014427	1425.739	0.297
Childless share	0.188000	0.080188	−0.107812	17180.744	199.700
Mean age at first birth	25.310561	25.637433	0.326872	16.000	1.710
Share of first births at age 30 or later	0.270062	0.332334	0.062272	15625.000	60.590
Rooms response at first birth (measurement hold)	0.804944	0.657205	−0.147739	35.735	0.780
Rooms gap, 3+ versus 1–2 children	0.367700	0.372240	0.004540	2958.515	0.061
Family ownership gap	0.167662	0.165878	−0.001783	14229.591	0.045
Ownership rate, ages 30–55	0.575472	0.563758	−0.011714	1207.846	0.166
Mean occupied rooms, ages 18–85	5.779970	6.075505	0.295534	11.973	1.046
Wealth / annual gross labor earnings	6.873100	5.454754	−1.418346	6.288	12.649
Annual bequest flow / aggregate wealth	0.008800	0.009078	0.000278	5165289.256	0.399
Old wealth-to-income p90/p50, ages 76–84	3.448111	2.068370	−1.379741	56.960	108.433
Exact current-contract loss					385.875

The table is an audit artifact, not a promoted calibration table. In particular, the first-birth rooms row remains under a PSID measurement hold. The unrounded row contributions sum exactly to the reported scalar; the small difference visible after summing the printed entries is rounding.

Table 3: Retained E5b benchmark: all free and externally fixed parameters

Parameter	Estimate	Search bound / restriction	Status
Annual discount factor β	0.991097	[0.94, 0.9995]	Estimated
Per-child expenditure shift Δ_1	0.012163	[0, 0.25]	Estimated
First-child expenditure jump Δ_0	0.023896	[0, 0.25]	Estimated
Value-of-child shifter ψ	0.585018	[−3, 3]	Estimated
First-birth logit scale κ_1	3.552006	[0.02, 50]	Estimated
Continuation logit scale κ_C	1.254006	[0.02, 50]	Estimated
Owner-service premium χ	1.038033	[0.1, 5]	Estimated
Housing-supply scale H_0	8.162391	[0.2, 80]	Estimated
Bequest intensity θ_0	0.168435	[0, 8]	Estimated
Bequest shift θ_1	0.050046	[0.02, 16]	Estimated; within 2% of lower bound
Childless-renter expenditure share α_0	0.733000	CEX, 2019–2023	Externally fixed
Tenure taste-shock scale	0	Author restriction	Externally fixed
Child dependence of bequest utility	0	Child-invariant bequest	Externally fixed

C Numerical illustration: parameters and replication

Table 4: Parameters of the generic transition illustration

Object	Value	Object	Value
Period length	5 years	Adult ages	20–79
Maturation delay	4 periods	Child survival per period	0.992
Household equivalents per birth ν	0.50	Initial housing-cost index	1.00
Housing-supply elasticity	0.65	Demand price elasticity	1.00
Child goods cost	0.72	Child space	0.22
Initial access wedge	0.24	Easier-access reduction	0.12
Initial value of children	1.16	Long-run value	0.94
Old-retention reduction	18%	Policy start	year 10

The pre-decline birth scale is chosen so the frozen demographic operator has dominant eigenvalue one at a housing-cost index of one. The initial adult distribution is a deterministic tilt of that stationary eigenvector, with a prime-age bulge and modestly smaller recent child cohorts. The supply scale is

chosen so the initial market also clears at one. These normalizations create a transparent sign experiment rather than an empirical calibration.

The two-panel steady-state figure is rebuilt from the repository root with

```
code/model/.venv/bin/python \  
code/model/tools/build_fertility_population_housing_transition_figures.py
```

The complete numerical-transition packet is rebuilt with

```
code/model/.venv/bin/python \  
code/model/tools/build_dynamic_population_housing_paper_illustrations.py
```

It writes aggregate paths, cohort distributions, the exact momentum-condition grid, figures, parameters, and all accounting residuals to `output/model/dynamic_population_housing_paper/`.

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